Supplementary Information

Validation–Deployment Alignment Shapes How Internal Evaluation Predicts External Generalization in Lithium Solid-Electrolyte Machine Learning

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This Supplementary Information follows the structure of the main-text Results: primary analysis, robustness controls, deployment-scenario refinement, risk analysis and provenance. All values are reproduced from the study's frozen analysis records; no numbers were re-derived for this document. Methods are described in the main text. The pooled elemental-property model is referred to by that name throughout. Internal-validation results obtained before the nested model-selection correction are not reported here; their chronology is summarized in Supplementary Note 27. The original pre-correction records are retained in the authors' internal Git history and are not included in the curated public release.

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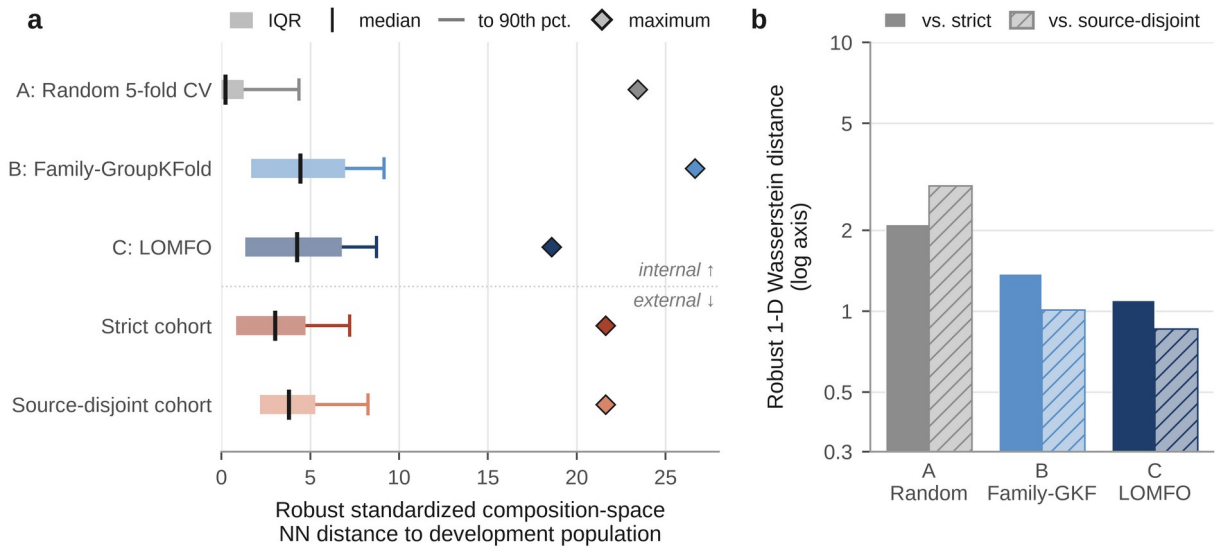
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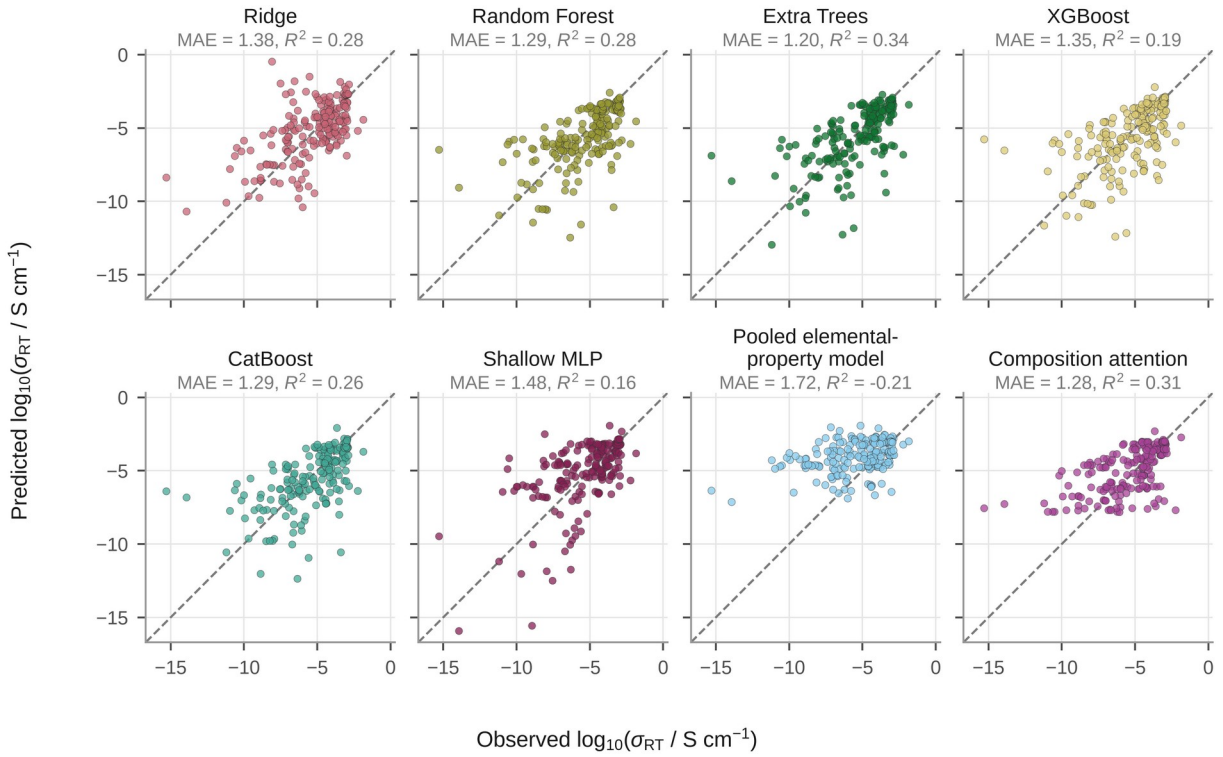
Part I. Additional post hoc robustness controls (added for this submission; each specified and hashed before computation): Note 30 – Common-support LOMFO sensitivity; Note 31 – Training-size-matched control for Family-GroupKFold; Note 32 – Strictly fold-local robust shift-distance correction; Note 33 –

Source/DOI-cluster bootstrap; Note 34 – __MISSING__-family sensitivity; Note 35 – Chemical-system-GroupKFold sensitivity (regime E)



Supplementary Fig. 1 | Robust composition-space shift alignment. a, Median-to-90th-percentile robust standardized nearest-neighbour distance with cohort maximum marked by a diamond. b, One-dimensional Wasserstein distance between each internal regime's held-out-distance distribution and the strict or DOI-source-disjoint external cohort. Removing near-degenerate fold-local dimensions eliminates the extreme scaling artefact without changing the qualitative alignment: LOMFO remains the closest distributional match.

Panel a summarizes the robust standardized nearest-neighbour distance distributions for the three internal validation regimes and the two external cohorts. Panel b compares the resulting one-dimensional Wasserstein distances to the strict and DOI-source-disjoint external distributions. These target-free diagnostics describe the type and magnitude of feature-space shift induced by each split; they are not used to select models or validation regimes.



Supplementary Fig. 2 | Observed versus predicted external conductivity for all eight models on the strict cohort. Dashed lines show identity; panel titles report external MAE and R². Shared axes make model calibration directly

comparable across panels, including the pooled elemental-property model's systematic scale/offset error (Supplementary Note 27). The plotted points use the corrected pooled elemental-property model predictions and the other seven models' unchanged frozen external predictions.

Parity plots use identical axis limits for all eight models on the strict external cohort (n = 204). The dashed diagonal is the identity line. The pooled elemental-property model shows negative R^2 and a real, positive rank correlation with the experimental values (Spearman $\rho = 0.39$) rather than the literal near-constant predictions an implementation bug previously produced (Supplementary Note 27); the leading tree and attention models retain substantial scatter but track the experimental ordering more closely. The plotted values are the corrected pooled elemental-property model predictions and the other seven models' original frozen external predictions.

Supplementary Table 1 | Robustness summary: strict, DOI-source-disjoint and 20–30 °C window cohorts.

Model	MAE (strict)	MAE (DOI-source-disjoint)	Δ (DOI-source-disjoint)	MAE (20–30 °C)	Δ (window)	Δ (window vs. strict)
Extremely randomized trees	1.2022	1.3673	0.1651	1.2113	0.0009	0.0091
Composition attention	1.2824	1.4036	0.1212	1.2715	−0.0019	−0.0108
Random forest	1.2861	1.4516	0.1655	1.3005	0.0009	0.0144
CatBoost	1.2916	1.4594	0.1678	1.3002	0.0009	0.0085
XGBoost	1.3492	1.4920	0.1427	1.3745	0.0009	0.0252
Ridge	1.3820	1.5030	0.1210	1.3712	0.0007	−0.0108
Shallow MLP	1.4831	1.6801	0.1970	1.4619	−0.0036	−0.0211
Pooled elemental-property model	1.7163	1.9447	0.2284	1.7018	0.0036	−0.0145

MAE in log10(S cm^{-1}). n = 204 (strict cohort), 153 (DOI-source-disjoint subset) and 197 (fixed 20–30 °C window). Δ (window) isolates the effect of narrowing the window while holding the formula subset fixed; Δ (window vs. strict) compares the same values with the full 204-row strict cohort. Per-model detail is given in Supplementary Notes 11 and 12.

Part A. External cohort and label compatibility

Supplementary Note 1. External dataset provenance and audit

Field	Value
Dataset	Hargreaves Lithium Solid Electrolyte Conductivity Database
Primary publication	Hargreaves, C. J. et al., npj Computational Materials 9, 9 (2023)
DOI	10.1038/s41524-022-00951-z
Authoritative source	https://pcwww.liv.ac.uk/~msd30/lmds/LiIonDatabase.csv
Retrieval date	2026-09-21
License	Academic use permitted with citation; commercial use requires separate University of Liverpool license
Local file SHA-256	805e157decb613e9d1ed651f4218c2ea69ed7d6fa2aa24b7b860485894570606
Row count (verified)	820, matches publisher abstract exactly
Unique sources (verified)	214, matches publisher abstract exactly
Censored rows	0
Temperature range	5.0–873.0 °C
RT-window (15–35 °C) discrepancy	Reconciled (see Supplementary Note 2).

Supplementary Note 2. Reconciliation of unique room-temperature composition counts

The 465-row 15–35 °C population is identical between our audit and the paper throughout. Re-running the paper's own published composition-normalization method (ElMD(...).pretty_formula) on those same 465 rows gives len(set(pretty_formula)) == 403, an exact bit-for-bit match.

Normalization	Unique count
Raw string (original method)	422
Whitespace-stripped	417
pymatgen reduced formula	406
ElMD pretty_formula (authors' method — exact match)	403

Supplementary Note 3. External cohort construction: stage counts

Stage	Rows
Raw Hargreaves	820
RT window [15,35]°C	465
Strict pre-aggregation (all overlap excluded)	213
Relaxed pre-aggregation (only exact-DOI excluded)	263
Strict composition-disjoint (post-aggregation)	204
Relaxed composition-disjoint (post-aggregation)	239

Exclusion breakdown (of 465 RT-window rows): 202 exact composition+DOI matches with OBELiX (43.4%), 50 composition-only matches (10.8%). Strict cohort ID set confirmed a proper subset of relaxed (204 $\subset$ 239). Aggregation rule: median log10(conductivity) per reduced formula among RT-window, non-excluded rows.

Supplementary Note 4. Source-only literature overlap between the strict cohort and OBELiX

The strict cohort is composition-disjoint from OBELiX by construction, but not DOI-source-disjoint: of the 213 pre-aggregation retained rows (91 unique source DOIs), 17 of those DOIs (53 rows, corresponding to 51 of the 204 aggregated strict-cohort compositions, 25.0%) also appear as a source DOI somewhere in OBELiX's 222 unique DOIs. This does not re-open any excluded row and does not change the 204/239 cohort counts. See Supplementary Note 11 for the DOI-source-disjoint sensitivity cohort this motivated.

Supplementary Note 5. Cross-database label compatibility: paired-formula detail

Lead statistic (dev-only, 449-row OBELiX development population). The OBELiX side is restricted to the 449-row development population (train+validation) only; the frozen 113-row test split's labels are never read for this comparison.

n paired formulas: 166 (135 exact composition-and-DOI match-only, 31 involving a composition-only match).

OBELiX side: 449 dev rows, 0 inequality-censored among the dev population, 365 exact formulas after pymatgen-reduction and per-formula median aggregation; 17 of those formulas have a within-OBELiX spread >1 log10(S/cm) across their own contributing rows (a pre-existing within-OBELiX phenomenon, not part of the cross-database comparison).

Discrepancy statistics: mean absolute discrepancy 0.292 (95% bootstrap CI [0.199, 0.393], n_boot=2000, seed 20260921); median absolute discrepancy 0.0071; RMSE 0.697; Spearman ρ 0.922 (95% CI [0.861, 0.965]); Pearson r 0.958; median signed difference (Hargreaves − OBELiX) 2.8×10^{-11} (i.e., zero to numerical precision); mean signed difference +0.062.

Stratified by overlap classification: exact composition-and-DOI match-only pairs (n=135) mean absolute discrepancy 0.186, median 3.1×10^{-9}; pairs involving a composition-only match (n=31) mean absolute

discrepancy 0.754, median 0.347. All 31 composition-only-match formulas are present in the 239-row relaxed cohort and none in the 204-row strict cohort (verified by direct membership check).

15/166 pairs (9.0%) differ by more than 1 log10 unit; 6/166 (3.6%) by more than 2 log10 units. The full paired-formula list is included in the release (artifacts/authoritative/phase2_label_compat_dev_only_full.json).

Descriptive comparison (not a formal test): the composition-only-match group's 0.75 mean absolute discrepancy is the same order of magnitude as each model's own SEEN-family external MAE (0.83–1.14, main-text Table 4) — the two are different quantities (measurement-vs-measurement disagreement vs. model-vs-measurement error) and this is descriptive context, not evidence that one explains the other.

SI-only sensitivity (full 599-row OBELiX file, including the frozen test split). This earlier version is retained only as a scope-sensitivity check, not this study's lead statistic, because it read labels from the frozen 113-row test split.

n paired formulas: 200 (165 exact composition-and-DOI match-only, 35 involving a composition-only match).

OBELiX side: 599 raw rows, 37 inequality-censored (excluded per the project data contract censoring policy), 468 exact formulas after pymatgen-reduction and per-formula median aggregation; 19 of those 468 formulas have a within-OBELiX spread >1 log10(S/cm).

Discrepancy statistics: mean absolute discrepancy 0.290 (95% bootstrap CI [0.206, 0.385], n_boot=2000, seed 20260921); median absolute discrepancy 0.0023; RMSE 0.694; Spearman ρ 0.922; Pearson r 0.956; median signed difference (Hargreaves − OBELiX) 4.6×10^{-11}; mean signed difference +0.045.

Stratified by overlap classification: exact composition-and-DOI match-only pairs (n=165) mean absolute discrepancy 0.184, median 4.9×10^{-10}; pairs involving a composition-only match (n=35) mean absolute discrepancy 0.787, median 0.347.

20/200 pairs (10.0%) differ by more than 1 log10 unit; 8/200 (4.0%) by more than 2 log10 units. The full paired-formula list, per-formula overlap classification and per-formula source-row counts are included in the release (artifacts/authoritative/n20_artifacts/reviewer_label_compatibility.json).

Metadata-field audit (OBELiX vs. Hargreaves, full detail): see the main text ('Cohort construction and label compatibility') .

Part B. Primary analysis: external transfer and validation-regime comparison

Supplementary Note 6. External transfer: strict and relaxed cohorts

run_count=1, authorized=true. The primary record was never rerun or overwritten. The pooled elemental-property model alone received one disclosed corrective refit under the same frozen protocol after its feature-lookup bug was fixed (Supplementary Note 27); its corrected predictions are combined with the seven unchanged models in a separately labelled, non-primary record.

Strict cohort (primary, n=204):

Rank	Model	MAE	RMSE	Spearman rho	Seed-dispersion (mean std)
1	Extremely randomized trees	1.2022	1.7787	0.6996	0.0654
2	Composition attention	1.2824	1.8163	0.6345	0.5505
3	Random forest	1.2861	1.8562	0.6749	0.0755
4	CatBoost	1.2916	1.8841	0.6740	0.3010
5	XGBoost	1.3492	1.9685	0.6618	0.4617
6	Ridge	1.3820	1.8644	0.5840	0.0000
7	Shallow MLP	1.4831	2.0064	0.5957	0.2302
8	Pooled elemental-property model	1.7163	2.4168	0.3929	0.1096

Relaxed cohort (companion sensitivity, n=239):

Model	MAE	RMSE	Spearman rho
Extremely randomized trees	1.1880	1.7970	0.6904
Composition attention	1.2354	1.7576	0.6444
Random forest	1.2465	1.8336	0.6727
CatBoost	1.2626	1.8833	0.6681
XGBoost	1.2999	1.9470	0.6654
Ridge	1.3421	1.8067	0.5733
Shallow MLP	1.4190	1.9458	0.5946
Pooled elemental-property model	1.6105	2.3001	0.3998

A pre-submission implementation audit found that the pooled elemental-property model's earlier near-constant mean-of-5-seeds prediction (constant value −4.167444 across all 204 strict-cohort rows, std 8.9×10^{-16}) was caused by a feature-lookup bug, not a genuine architectural collapse; see Supplementary Note 27 for the correction and its effect on every dependent statistic. The relaxed-cohort row above uses the corrected representation; full propagation of the correction through every remaining table and figure in this document is tracked in Supplementary Note 27; the corrected values are in artifacts/authoritative/.

Supplementary Note 7. External performance context

Pure post-hoc recomputation from the authoritative external-transfer strict-cohort predictions (n=204). This analysis did not rerun the external transfer; it reused the authoritative external predictions: seven original frozen models plus the disclosed corrected pooled elemental-property model. MAE/RMSE were independently recomputed and cross-checked against the authoritative external-transfer mae/rmse fields to within 1e-6 for every model (assertion in the script). Naive baselines predict the OBELiX development-population's own mean (−4.8182) / median (−4.0841) log10(σRT) for every external row — never the external cohort's own mean/median, which would let external labels leak into the baseline itself. Mean-baseline MAE = 1.6701; median-baseline MAE = 1.7429. Skill score = 1 − MAE_model / MAE_baseline (positive = model beats the naive baseline).

Model	MAE	RMSE	R2	Pearson r	Spearman rho	Skill vs. mean baseline	Skill vs. median baseline
Extremely randomized trees	1.2022	1.7787	0.3425	0.6465	0.6996	0.2802	0.3102
Composition attention	1.2824	1.8163	0.3144	0.6030	0.6345	0.2322	0.2642
Random forest	1.2861	1.8562	0.2840	0.6111	0.6749	0.2299	0.2621
CatBoost	1.2916	1.8841	0.2623	0.5953	0.6740	0.2266	0.2589
XGBoost	1.3492	1.9685	0.1948	0.5764	0.6618	0.1921	0.2259
Ridge	1.3820	1.8644	0.2777	0.6049	0.5840	0.1725	0.2071
Shallow MLP	1.4831	2.0064	0.1634	0.5939	0.5957	0.1120	0.1491
Pooled elemental-property model (pooled elemental-property model)	1.7163	2.4168	−0.2138	0.3831	0.3929	−0.0277	0.0152

The pooled elemental-property model shows negative R^2 (−0.21, a systematic scale/offset error) despite real, positive Pearson and Spearman correlations with the external labels (0.38 and 0.39 respectively) — not the undefined, zero-variance-constant behavior reported in an earlier analysis affected by a since-corrected feature-pipeline bug (Supplementary Note 27). It remains the only model with a negative skill score against the mean baseline (it beats the weaker median baseline only marginally, skill = 0.0152). All other seven models beat both naive baselines by a comfortable margin (skill 0.11–0.31).

Supplementary Note 8. External uncertainty

Row-paired bootstrap, n_boot=2000, seed=20260921.

Strict cohort (n=204, primary):

Model	MAE	95% CI	P(rank 1)
CatBoost	1.2916	[1.1203, 1.4878]	0.0115
Composition attention	1.2824	[1.1038, 1.4579]	0.1770
Extremely randomized trees	1.2022	[1.0354, 1.3912]	0.7855
Pooled elemental-property model	1.7163	[1.4882, 1.9510]	0.0000
Random forest	1.2861	[1.1126, 1.4793]	0.0180
Ridge	1.3820	[1.2172, 1.5564]	0.0060
Shallow MLP	1.4831	[1.2982, 1.6648]	0.0000
XGBoost	1.3492	[1.1640, 1.5497]	0.0020

DOI-source-disjoint cohort (n=153, companion):

Model	MAE	95% CI	P(rank 1)
CatBoost	1.4594	[1.2345, 1.6960]	0.0215
Composition attention	1.4036	[1.1904, 1.6237]	0.3440
Extremely randomized trees	1.3673	[1.1469, 1.5911]	0.5535
Pooled elemental-property model	1.9447	[1.6613, 2.2193]	0.0000
Random forest	1.4516	[1.2414, 1.6727]	0.0230
Ridge	1.5030	[1.2938, 1.7282]	0.0415
Shallow MLP	1.6801	[1.4556, 1.9020]	0.0000
XGBoost	1.4920	[1.2680, 1.7324]	0.0165

The full pairwise win-rate matrix (descriptive only; not used for any decision) is included in the release (artifacts/authoritative/n15_artifacts/n15_external_uncertainty.json).

Supplementary Note 9. Nested internal validation: per-model, per-regime results

Model	Regime	Pooled MAE	Macro-family MAE	RMSE	Spearman rho	n scored
Ridge	A: random five-fold CV	1.4103	3.0041	2.8174	0.6900	449
Ridge	B: Family-GroupKFold	143.1289	30.1480	727.7077	0.2405	449
Ridge	C: LOMFO	2.4170	2.1141	2.9620	0.3370	396
Random forest	A: random five-fold CV	0.8256	2.0918	1.3880	0.8695	449
Random forest	B: Family-GroupKFold	2.0300	2.5033	2.5735	0.4679	449
Random forest	C: LOMFO	2.1885	2.0672	2.6980	0.3772	396
Extremely randomized trees	A: random five-fold CV	0.7497	1.8223	1.2551	0.8864	449
Extremely randomized trees	B: Family-GroupKFold	1.8202	2.2378	2.3027	0.5317	449
Extremely randomized trees	C: LOMFO	2.0035	1.9083	2.4842	0.4721	396

Model	Regime	Pooled MAE	Macro-family MAE	RMSE	Spearman rho	n scored
XGBoost	A: random five-fold CV	0.8120	2.0613	1.3537	0.8602	449
XGBoost	B: Family-GroupKFold	1.4548	2.4643	1.9830	0.6229	449
XGBoost	C: LOMFO	1.8582	1.7358	2.3630	0.4861	396
CatBoost	A: random five-fold CV	0.8017	1.8822	1.3717	0.8747	449
CatBoost	B: Family-GroupKFold	1.4867	2.2184	1.9866	0.5974	449
CatBoost	C: LOMFO	1.5499	1.6067	1.9708	0.5749	396
Shallow MLP	A: random five-fold CV	1.0293	2.3396	1.7405	0.7773	449
Shallow MLP	B: Family-GroupKFold	969.4243	182.2681	4997.1963	0.3194	449
Shallow MLP	C: LOMFO	2.0039	2.1221	2.6147	0.3101	396
Pooled elemental-property model	A: random five-fold CV	1.5511	2.8366	2.4011	0.4160	449
Pooled elemental-property model	B: Family-GroupKFold	2.7024	3.1570	3.4537	0.1636	449
Pooled elemental-property model	C: LOMFO	2.2180	2.3044	2.8307	0.2601	396
Composition attention	A: random five-fold CV	1.0478	2.2760	1.7459	0.7738	449
Composition attention	B: Family-GroupKFold	1.8504	2.7541	2.5934	0.2800	449
Composition attention	C: LOMFO	1.4507	1.9179	2.0781	0.4857	396

Exact Family-GroupKFold (regime B) fold composition (GroupKFold(5) grouped by the harmonized family_filled label, deterministic given input row order; no shuffling parameter — reproduced by sse_transfer.evaluation.internal_validation.family_group_kfold_splits with seed=20260921, which controls the row-order the pool is loaded in, not a random fold assignment):

Fold	n held out	Families held out
0	213	LGPS, NASICON, garnet-like, hexaoxometalates, oxides, perovskites
1	29	__MISSING__, boron nitride, chlorides, nitridosilicates, pyrophosphate, zirconates
2	141	argyrodites, copper pyroborate, garnet, halides, layered, molybdates
3	31	antiperovskite, boracites, fluorides, nitrides, phosphates, thio-LISICON
4	35	Chalcogenidotetrelates, LISICON, Tavorite, hydrides, sulfides, thio-phosphate

Fold 0 alone holds out NASICON (n=130, the single largest family) together with five smaller families, giving it by far the largest test partition (213/449, 47.4%) and making it the fold responsible for the ridge/shallow-MLP catastrophic blowups reported in the main text (Fig. 5) — GroupKFold's greedy group-balancing algorithm does not guarantee equal-size folds when group sizes are as skewed as this study's family distribution (NASICON n=130 vs. a median family size of 5).

Supplementary Note 10. Validation-regime utility (nested)

Regime	Median | MAE gap |	Spearman rho	Kendall tau	Top-model regret
A: random five-fold CV	0.4531	0.6429	0.5000	0.0000
B: Family-GroupKFold	0.6810	0.5476	0.3571	0.1471
C: LOMFO	0.5149	0.5714	0.4286	0.0802

H1/H2 verdicts: all six (2 H1 + 4 H2) comparisons NOT SUPPORTED. Random 5-fold CV (A) wins all four statistics by point estimate, but the B/C-minus-A rank-agreement 95% CIs now include zero after the pooled elemental-property model's representation was corrected (Supplementary Note 27) -- only the B-minus-A regret difference remains significant (Table 3 / main text). See Supplementary Note 14 for the six-model stable-sensitivity recomputation of this same table.

Part C. Robustness controls

Supplementary Note 11. DOI-source-disjoint external sensitivity

Strict cohort n=204; DOI-source-disjoint n=153 (51 compositions excluded, 25.0%). OBELiX DOI set size (after |-splitting multi-DOI cells): 225.

Model	MAE (strict)	RMSE (strict)	Spearman (strict)	MAE (DOI-source-disjoint)	RMSE (DOI-source-disjoint)	R^2 (DOI-source-disjoint)	Spearman (DOI-source-disjoint)	n	MAE delta
Ridge	1.3820	1.8644	0.5840	1.5030	2.0254	0.2403	0.5776	153	+0.1210
Random forest	1.2861	1.8562	0.6749	1.4516	2.0318	0.2355	0.6419	153	+0.1655
Extremely randomized trees	1.2022	1.7787	0.6996	1.3673	1.9532	0.2935	0.6615	153	+0.1651
XGBoost	1.3492	1.9685	0.6618	1.4920	2.1342	0.1565	0.6436	153	+0.1427
CatBoost	1.2916	1.8841	0.6740	1.4594	2.0803	0.1986	0.6411	153	+0.1678
Shallow MLP	1.4831	2.0064	0.5957	1.6801	2.2116	0.0943	0.5535	153	+0.1970
Pooled elemental-property model	1.7163	2.4168	0.3929	1.9447	2.6316	-0.2824	0.3728	153	+0.2284
Composition attention	1.2824	1.8163	0.6345	1.4036	1.9568	0.2909	0.6042	153	+0.1212

For comparison, an earlier, simpler method reported 51/204 using a simpler per-composition-only method; this analysis independently reproduces 51/204 exactly using the full DOI-set method above.

Supplementary Note 12. Fixed 20–30 °C temperature-window sensitivity

n under [20, 30] °C = **197** (of 204 strict-cohort formulas under [15, 35] °C).

Model	MAE (20-30)	MAE (15-35, same subset)	n	Delta (window only)	Delta (vs. full strict n=204)
Ridge	1.3712	1.3705	197	+0.0007	-0.0108
Random forest	1.3005	1.2996	197	+0.0009	+0.0144
Extremely randomized trees	1.2113	1.2104	197	+0.0009	+0.0091
XGBoost	1.3745	1.3735	197	+0.0009	+0.0252
CatBoost	1.3002	1.2993	197	+0.0009	+0.0085
Shallow MLP	1.4619	1.4655	197	-0.0036	-0.0211
Pooled elemental-property model	1.7018	1.6982	197	+0.0036	-0.0145
Composition attention	1.2715	1.2735	197	-0.0019	-0.0108

"Delta (window only)" isolates the effect of narrowing the temperature window while holding the formula subset fixed (n=197 in both columns); "Delta (vs. full strict n=204)" additionally reflects the smaller formula subset. All window-only deltas are below 0.004 absolute.

Supplementary Note 13. Constant baselines and repeated-measurement variability

OBELiX-development mean(y) = -4.8182, median(y) = -4.0841.

Cohort	n	Mean-predictor MAE	Mean-predictor RMSE	Mean-predictor R^2	Median-predictor MAE	Median-predictor RMSE	Median-predictor R^2
strict cohort	204	1.6701	2.2508	-0.0528	1.7429	2.5191	-0.3187
DOI-source-disjoint cohort	153	1.8018	2.4452	-0.1073	1.9676	2.7632	-0.4140

Within-composition raw repeated-measurement variability (RT-window, valid-target rows, n_groups_with_repeats=37, min_repeats_threshold=2, median n_repeats=2.0 [IQR 2.0, 3.0]): median log10-range 0.4949 (IQR [0.1548, 1.2944]), pooled MAD 0.1811. Labeled as: empirical measurement/temperature-protocol/synthesis heterogeneity across independent literature reports of a nominally identical composition — NOT an irreducible noise floor (main text, 'External performance in context' and Limitations).

Supplementary Note 14. Six-model stable-architecture sensitivity

Excluded models: ridge, Shallow MLP. Retained: Random forest, Extremely randomized trees, xgboost, catboost, Pooled elemental-property model, Composition attention. Primary 8-model nested regime-utility result (Supplementary Note 10) is unmodified by this analysis.

Regime	Median |MAE gap|	Spearman rho	Kendall tau	Top-model regret
A	0.4565	0.4857	0.3333	0.0000
B	0.5930	0.0857	0.0667	0.1471
C	0.5053	0.3714	0.3333	0.0802

Pairwise bootstrap CIs (n_boot=2000, seed=20260921), 6-model:

Pair	Statistic	Point delta	95% CI
B − A	median |gap|	0.1365	[-0.1905, 0.4466]
B − A	Spearman ρ	-0.4000	[-0.5714, -0.1143]
B − A	Kendall τ	-0.2667	[-0.5333, 0.0000]
B − A	regret	0.1471	[0.0337, 0.2638]
C − A	median |gap|	0.0488	[-0.2975, 0.3751]
C − A	Spearman ρ	-0.1143	[-0.8571, 0.4000]
C − A	Kendall τ	0.0000	[-0.8000, 0.2667]
C − A	regret	0.0802	[-0.0997, 0.2557]
C − B	median |gap|	-0.0877	[-0.1853, 0.0207]
C − B	Spearman ρ	0.2857	[-0.4571, 0.5143]
C − B	Kendall τ	0.2667	[-0.5333, 0.5333]
C − B	regret	-0.0669	[-0.2641, 0.1311]

H1 (family-aware regime's median |MAE gap| smaller than random CV's): B − A **NOT SUPPORTED**, C − A **NOT SUPPORTED**. H2 (family-aware regime's rank agreement higher than random CV's): B − A Spearman ρ, B − A Kendall τ, C − A Spearman ρ, C − A Kendall τ all **NOT SUPPORTED**.

Supplementary Note 15. Exact-duplicate audit and formula-GroupKFold control

The duplicate-formula diagnostic used the archived source records and the same one-process-per-model, OOM-avoidance convention used throughout the nested-CV and sensitivity analyses.

Duplicate audit. 52 of 449 OBELiX development-population formulas (136 rows, 30.3%) are exact pymatgen-reduced-formula duplicates. Under the fixed random-5-fold-CV partition (KFold(5, shuffle=True, random_state= 20260921) over row position 0..448, identical to every other use of this partition in this study), 45 of the 52 duplicate-formula groups (122 of 136 rows) have member rows in more than one outer fold; 7 groups (14 rows) are entirely within one fold. Full per-formula fold-membership detail is included in the release (artifacts/authoritative/n20_artifacts/reviewer_formula_duplicate_audit.json). Separately (descriptive only, not leakage): 63 chemical systems (334 rows) share an element set across more than one distinct stoichiometry (same record).

Formula-GroupKFold diagnostic regime — full per-model pooled MAE (vs. the nested internal validation's primary random-5-fold-CV pooled MAE):

Model	Random 5-fold CV	Formula-GroupKFold	Δ	Δ %
Ridge	1.4103	1.3676	−0.0427	−3.0%
Random Forest	0.8256	0.8459	+0.0203	+2.5%
Extra Trees	0.7497	0.7563	+0.0066	+0.9%
XGBoost	0.8120	0.7954	−0.0166	−2.0%
CatBoost	0.8017	0.7869	−0.0148	−1.8%
Shallow MLP	1.0293	1.1214	+0.0921	+8.9%
Pooled elemental-property model	1.5511	1.5134	−0.0377	−2.4%
Composition attention	1.0478	1.2979	+0.2501	+23.9%

Every fold used the same nested per-fold hyperparameter-tuning protocol as the nested internal validation (inner 80/20 split of that fold's own outer-train partition; same search spaces, model families, trial budgets, seed=20260921). No blowup comparable to the Family-GroupKFold family-exclusion blowups (main-text Fig. 5; 101.5–941.8×) occurred under this regime for any model; the largest relative change (composition attention,

+23.9%) is nearly two orders of magnitude smaller. Model ranking by pooled MAE is essentially preserved (extra trees, catboost, and xgboost remain the three lowest-MAE models; the pooled elemental-property model remains highest). Full per-fold hyperparameters and per-family breakdowns are included in the release (artifacts/authoritative/n20b_formula_group_kfold_corrected.json).

Verdict. The exact-formula leakage concern is real (45/52 duplicate-formula groups cross folds) but does not explain this study's central negative result: random-CV's more favorable (lower, closer-to-external) MAE relative to the family-aware regimes persists essentially unchanged under a regime that removes this specific leakage mechanism entirely.

Supplementary Note 16. Fold-randomness of random five-fold CV (ten seeds)

10 deterministic seeds (20260921, identical to every other analysis's regime-A seed, plus 20260922-20260930), 5 outer folds each, identical nested per-fold Optuna tuning protocol to the nested internal validation (Methods/Supplementary Note 9). Only regime A (random 5-fold CV) was repeated; regimes B (Family-GroupKFold) and C (LOMFO) are deterministic given the fixed family taxonomy and reuse their already-frozen nested internal-validation values. Total compute: 16,781s (4.66h) across 80 (seed, model) nested-tuning runs. This analysis did not rerun the external transfer; it reused the authoritative external predictions: seven original frozen models plus the disclosed corrected pooled elemental-property model. The frozen 113-row OBELiX test split was not opened.

Primary eight-model comparison.

Per-model internal (regime A) MAE across 10 seeds:

Model	Mean	Median	SD	Range across 10 seeds	Original seed
Ridge	1.3725	1.3604	0.1117	[1.2452, 1.6485]	1.4103
Random forest	0.8316	0.8312	0.0181	[0.8039, 0.8600]	0.8256
Extremely randomized trees	0.7697	0.7651	0.0208	[0.7435, 0.8090]	0.7497
XGBoost	0.8244	0.8114	0.0307	[0.7931, 0.8777]	0.8120
CatBoost	0.7981	0.7957	0.0280	[0.7618, 0.8515]	0.8017
Shallow MLP	1.1010	1.1163	0.0397	[1.0293, 1.1607]	1.0293
Pooled elemental-property model	1.6273	1.5938	0.1387	[1.4776, 1.8772]	1.5511
Composition attention	1.3529	1.4152	0.2617	[0.9932, 1.6856]	1.0478

"Range across 10 seeds" is the true observed [min, max] over the 10 seed values, not a percentile-based CI (see the regime-utility-statistics tables below for the same relabeling and rationale).

Regime-utility statistics, distribution across 10 seeds vs. fixed B/C:

Statistic	A mean	A median	A SD	A range across 10 seeds	B (fixed)	C (fixed)	Verdict
median |MAE gap|	0.4094	0.4030	0.0233	[0.3789, 0.4531]	0.6810	0.5149	Robust (A)
Spearman rho	0.5952	0.5714	0.1077	[0.4524, 0.7619]	0.5476	0.5714	Not robust (range contains B and C)
Kendall tau	0.4643	0.4643	0.0772	[0.3571, 0.5714]	0.3571	0.4286	Not robust (range contains B and C)
Top-model regret	0.0237	0.0000	0.0517	[0.0000, 0.1471]	0.1471	0.0802	Not robust (CI overlaps C)

Verdict: 2/4 statistics fold-randomness-robust for A under a strict no-overlap standard (median |MAE gap| and top-model regret); rank agreement (Spearman rho, Kendall tau), previously reported as robust, is not -- after the pooled elemental-property model's representation was corrected (Supplementary Note 27), the 10-seed range for both rank-agreement statistics now contains the fixed B and C values. "A range across 10 seeds" is the raw

observed [min, max] over the 10 seed values, not a percentile-based confidence interval; an earlier version of this table reported the 2.5th/97.5th percentile of these same 10 points, mislabeled as a "95% interval" — the percentile and the true range differ only slightly here but the percentile label is not statistically justified at n=10, so this table now reports the true range.

Six-model stable-architecture sensitivity (excludes ridge, Shallow MLP).

Per-model internal (regime A) MAE across 10 seeds: identical values to the primary comparison above, restricted to the six retained models (no recomputation of the underlying fits — the same 10x5 nested-CV runs are reused; ridge/Shallow MLP's rows are simply excluded from this subset's summary statistics).

Regime-utility statistics, distribution across 10 seeds vs. fixed B/C (six-model):

Statistic	A mean	A median	A SD	A range across 10 seeds	B (fixed)	C (fixed)	Verdict
median | MAE gap |	0.4435	0.4465	0.0185	[0.4144, 0.4674]	0.5930	0.5053	Robust (A)
Spearman rho	0.4400	0.4857	0.1136	[0.2000, 0.6000]	0.0857	0.3714	Not robust (range contains C)
Kendall tau	0.3067	0.3333	0.1052	[0.0667, 0.4667]	0.0667	0.3333	Not robust (range contains B, C)
Top-model regret	0.0237	0.0000	0.0517	[0.0000, 0.1471]	0.1471	0.0802	Not robust (CI overlaps C)

Verdict: 1/4 statistics fold-randomness-robust for A under a strict no-overlap standard. As above, "A range across 10 seeds" is the true observed [min, max], not a percentile-based CI. After the pooled elemental-property model's representation was corrected (Supplementary Note 27), the median-gap statistic's single-seed result now favours random CV in this subset (reversing the earlier family-aware-favouring result) and is itself robust to fold-randomness: under repeated sampling, random CV's median gap never rises into the family-aware regimes' range. Rank agreement (Spearman rho, Kendall tau) is no longer robust in either direction -- LOMFO's fixed value now falls inside random CV's ten-seed range for both statistics, and Family-GroupKFold's falls inside for Kendall tau -- and top-model regret remains not robust, as in the eight-model comparison.

Interpretation.

Both comparisons' central qualitative conclusion (H1/H2 NOT SUPPORTED for either family-aware regime) is unaffected: nowhere does a family-aware regime's fixed value fall outside random CV's repeated-sampling distribution in the direction that would newly support H1/H2. What changes is the strength of the claim that CAN be made about random CV's own advantage: "wins on N of 4 predeclared statistics" is not a single fixed fact but a distribution-dependent count once fold-randomness is taken into account, and this section reports that count honestly for both comparisons rather than treating the original single-seed realization as definitive.

Part D. Deployment-scenario refinement

Supplementary Note 17. Family crosswalk and SEEN/UNSEEN classification

Provenance and normalization of family labels. OBELiX's Family column and Hargreaves' family column are each author-curated in their respective source databases (refs. S1, S2) — neither was assigned or re-labeled by this study. OBELiX's raw Family strings were harmonized once, by human review, in an earlier project phase (frozen, version-pinned harmonization module, version 1.0): 4 raw-string pairs were merged as spelling/casing/pluralization variants of the same physical family ("Halide"→"halides", "argyrodite"→"argyrodites", "cholorides" [a transcription typo]→"chlorides", "nitride"→"nitrides"; canonical form chosen by majority vote among the affected rows), and "LISICON" vs. "NASICON" was explicitly investigated and **rejected** as a harmonization candidate despite a small raw edit distance (2 of 7 characters) — these are two chemically distinct, well-known SSE families (Lithium vs. Sodium Super Ionic CONductor), not a spelling variant. Every other raw Family value passes through unchanged; none is guessed or invented. Rows with no Family value at all are filled with an explicit __MISSING__ sentinel (18 of 449 development-

population rows; the archived pre-correction record) rather than being dropped or imputed from any other column — __MISSING__ is then treated as its own group for Family-GroupKFold/LOMFO purposes (limitation: a __MISSING__ "family" mixes genuinely unrelated compositions that merely lack a label, so any regime statistic computed specifically on that group, e.g. the main-text SEEN/UNSEEN split treats __MISSING__-labeled OBELiX rows identically to any other absent match, is descriptive only, not a claim that __MISSING__ compositions are structurally related to each other). The Hargreaves→OBELiX crosswalk used for the SEEN/UNSEEN/AMBIGUOUS classification below is a separate, exact-string-match-only step (no fuzzy matching, no manual per-formula override) performed for this study by a deterministic script; its full decision table is immediately below.

Full crosswalk table (raw Hargreaves family label → matched OBELiX family, status, n strict-cohort formulas):

Hargreaves family	Matched OBELiX family	Status	n (strict cohort)
Anti-Perovskite	antiperovskite	SEEN	5
Argyrodite	argyrodites	SEEN	8
Garnet	garnet	SEEN	4
LISICON	LISICON	SEEN	6
NASICON	NASICON	SEEN	61
Perovskite	perovskites	SEEN	45
Thio-LISICON	thio-LISICON	SEEN	9
Glass	—	no OBELiX match	31
Glass-Ceramic	—	no OBELiX match	5
Lysonite	—	no OBELiX match	0
Olivine	—	no OBELiX match	6
Phenakite	—	no OBELiX match	3
Rocksalt	—	no OBELiX match	6
Zircon	—	no OBELiX match (intentionally not merged with "zirconates")	4
Other	—	AMBIGUOUS (non-informative label)	11

Totals: SEEN 138, UNSEEN 55 (sum of the seven no OBELiX match rows), AMBIGUOUS 11; 138+55+11=204. Every strict-cohort formula's raw family value agreed across all of its contributing pre-aggregation Hargreaves rows (zero multi-source family disagreement cases). Hargreaves' separate, coarser ChemicalFamily column (Halides/Oxides/Sulphides/Other — an anion-based grouping structurally different from OBELiX's structural Family column) was recorded per formula but not used for the SEEN/UNSEEN classification; it is retained in the archived family-audit record.

Per-model, per-subset metrics (MAE, RMSE, Spearman ρ and n, with 95% bootstrap CIs where $n \geq 10$) are included in the release (artifacts/authoritative/n20_artifacts/reviewer_family_audit.json); headline MAE/CI values are in main-text Table 4.

Regime-anticipation statistics (median |gap|, Spearman ρ, Kendall τ and regret, computed with the same routine as the primary analysis and applied to the nested internal-CV MAE against the SEEN and UNSEEN external-MAE subsets separately) are included in the same record; headline values are quoted in the main text ('Deployment scenario').

Supplementary Note 18. Material-class refinement of the UNSEEN subset: glass/glass-ceramic versus crystalline

This analysis did not rerun the external transfer and retrained no model; it reused the authoritative external predictions: seven original frozen models plus the disclosed corrected pooled elemental-property model. The

official 113-row OBELiX test split was not used for fitting, tuning, internal validation, external predictive-performance evaluation or any reported model-performance estimate. A historical post hoc label-compatibility audit queried the full OBELiX table, including test labels; the final lead compatibility analysis was rerun on development rows only.

Pre-declaration. A subset-size-dependent statistical-treatment rule and a three-branch conditional narrative-framing rule were written to a specification file and SHA-256-hashed **before** any subset MAE, regime gap, or AD-score distribution was computed: n≥30 for a full regime-ranking comparison with bootstrap CIs; n≥20 for MAE with a bootstrap CI only; below n=20, descriptive statistics only, with no regime-ranking statistic computed. The framing rule's branch 3 (crystalline-unseen below n=20, "report descriptively, state unresolved") is checked first, mechanically, before branches 1/2 are evaluated for the glass subset — the crystalline subset is never given a regime-ranking verdict regardless of what its raw numbers would have shown, and no such verdict was computed to be discarded.

Split. Crosswalking each of the 55 UNSEEN-family strict-cohort formulas' raw (non-harmonized) family label against {Glass, Glass-Ceramic} gives **36 glass/glass-ceramic** and **19 crystalline-but-unseen-family** formulas; the crystalline group's raw family breakdown (Zircon, Phenakite, Olivine, Rocksalt, and related small families) is included in the release (artifacts/authoritative/phase1_glass_split_corrected.json).

Per-subset × model metrics (per_model_subset_metrics): full MAE, RMSE, R^2, Spearman ρ, and (where n≥20) bootstrap CI for all eight models on SEEN (n=138), UNSEEN-all (n=55, unchanged from Supplementary Note 17), UNSEEN-glass (n=36), and UNSEEN-crystalline (n=19, descriptive only: mean MAE 2.67, range 1.86–4.48). Ranked by MAE, UNSEEN-crystalline is harder than UNSEEN-glass for five of eight models and easier for three (ridge, extra trees, random forest) — reported descriptively, not as a regime-ranking result.

Regime-utility, glass subset (n=36, point estimates). Source: regime_utility_by_subset.UNSEEN-glass.

Regime	Median |MAE gap|	Spearman ρ	Kendall τ	Top-model regret
A: random 5-fold CV	1.1298	0.1429	0.2143	0.0761
B: Family-GroupKFold	0.4094	0.3095	0.2143	0.1297
C: LOMFO	0.1539	0.3810	0.2857	0.0000
D: formula-GroupKFold	1.0848	0.1429	0.2143	0.0761

Glass-subset bootstrap CIs on the between-regime delta (seed 20260921, n_boot=2000, reusing the external transfer's frozen strict-cohort residuals restricted to the 36 glass formulas):

Pair	Statistic	Point delta	95% CI	Excludes zero?
B-A	median |MAE gap|	-0.720	[-0.828, -0.371]	Yes (favors B)
B-A	Spearman ρ	0.167	[-0.357, 0.476]	No
B-A	Kendall τ	0.000	[-0.429, 0.429]	No
B-A	Top-model regret	0.054	[-0.283, 0.391]	No
C-A	median |MAE gap|	-0.976	[-1.086, -0.373]	Yes (favors C)
C-A	Spearman ρ	0.238	[-0.619, 0.643]	No
C-A	Kendall τ	0.071	[-0.643, 0.500]	No
C-A	Top-model regret	-0.076	[-0.795, 0.650]	No
D-A	median |MAE gap|	-0.045	[-0.088, 0.027]	No
D-A	Spearman ρ	0.000	[0.000, 0.000]	No
D-A	Kendall τ	0.000	[0.000, 0.000]	No
D-A	Top-model regret	0.000	[0.000, 0.000]	No

Only the median-|MAE-gap| deltas for B-A and C-A exclude zero; every rank-agreement and regret delta, for every pair, has a 95% CI that includes zero. This is the quantitative basis for stating the glass-subset finding as

"anticipates error magnitude, not model selection" (main-text 'Deployment scenario') rather than a broader family-aware-CV advantage. The D−A (formula-GroupKFold) deltas are all CI-consistent with zero, confirming this exact-formula-disjoint control tracks random CV on the glass subset too, not the family-aware regimes.

Composition-space AD-score distribution, all four subsets (ad_score_distribution_by_subset): SEEN median 1.58 (n=138), UNSEEN-all median 5.26 (n=55), UNSEEN-glass median 5.29 (n=36), UNSEEN-crystalline median 3.83 but mean 6.59 / max 21.63 (n=19, one clear outlier) — reported descriptively; no new risk-stratification claim is made from this breakdown beyond the existing AD result (main-text 'Risk stratification'; Supplementary Note 23).

Chemical-system overlap (descriptive). Only 8 of the 36 unseen-glass formulas (22.2%) share even a chemical system (exact pymatgen element set) with any of the 449 OBELiX development-population rows — a materially weaker overlap condition than chemistry-family membership (e.g. an oxide glass can share every element with a crystalline OBELiX perovskite while having no structural family match at all). This is offered as descriptive context for why the glass subset is hard, not as a new leakage claim; the frozen 113-row test split was never consulted for this count.

Framing-rule verdict (mechanical). Source: stage0_framing_rule_verdict. branch 3 fires for crystalline-unseen (n=19<20; no regime-ranking verdict exists for this subset by construction); branch 2 fires for the glass subset (the smaller-median-gap pattern holds only for glass, not for the untested crystalline case). Resulting framing (mechanical rule output): "unseen material class (amorphous), not chemistry novelty." Branch 1 (a general unseen-chemistry claim) cannot fire, because the crystalline-unseen regime comparison it would require was pre-declared untestable at this sample size — this is a genuine, disclosed evidentiary gap, not a favorable result that was computed and omitted. The final manuscript (main-text 'Deployment scenario', Discussion, Limitations) does not use this label as its conclusion; it instead reports the more conservative evidentiary interpretation — the supported effect is specific to the unseen glass/glass-ceramic subset, and whether it generalizes to unseen crystalline chemistry remains unresolved — because branch 1 could not fire.

Provenance chain. phase1_predeclared.json (Stage 0, written and hashed before any Stage 1 statistic) → phase1_glass_split.json (Stage 1: split, per-subset metrics, point-estimate regime utility, AD distributions, chemical-system overlap, framing verdict) → phase2_glass_subset_regime_cis.json (bootstrap CIs on the glass-subset between-regime deltas, completing Stage 1's point estimates with formal statistical support). All three scripts assert n_dev_rows == 449 / equivalent frozen-population checks before computing anything, and none opens the frozen 113-row OBELiX test split.

Part E. Extrapolation failure and deployment-shift diagnostic

Supplementary Note 19. Deployment-shift alignment

No target y/log_target value was read anywhere in this analysis.

Standardized composition-space NN-distance distribution summary:

Regime/cohort	n	Median	Q25	Q75	IQR	Q90	Max	Mean
A	449	0.2135	0.0000	1.2541	1.2541	4.3568	23.4227	1.4542
B	449	4.7476	1.7507	7.5591	5.8084	10.0632	94559.5361	3243.5631
C	396	4.2561	1.3463	6.7701	5.4238	8.7324	18.5886	4.3982
strict cohort	204	3.0114	0.8224	4.7250	3.9027	7.2101	21.6266	3.4989
DOI-source-disjoint cohort	153	3.7911	2.1778	5.2849	3.1071	8.2503	21.6266	4.3444

Family novelty fraction (held-out row's family absent from its fold's train set): A 0.0312 (n=449), B 1.0000 (n=449), C 1.0000 (n=396).

1-D Wasserstein distance, internal regime vs. external cohort:

Regime	vs. strict	vs. DOI-source-disjoint
A	2.0998	2.9267
B	3240.1031	3239.5373
C	1.0962	0.8584

Regime B's max standardized NN distance (94559.5) is far outside its own q90 (10.06); one held-out fold's excluded family sits enormously far, in standardized composition-feature space, from that fold's training pool — the same family-removal mechanism identified independently in Supplementary Note 9 and main-text Fig. 5. This dominates regime B's Wasserstein distance (integrates over the full distribution, not just central quantiles).

Forensic sanity check of this maximum (post hoc diagnostic; does not alter any reported number). The maximum occurs in fold 0, for held-out row vtr (OBELiX ID; reduced formula Li6.0Fe6.0Sb6.0P18.0O72.0, family NASICON). Fold 0's training pool (n=236, after excluding the LGPS, NASICON, garnet-like, hexaoxometalates, oxides, and perovskites families) retains only 5/236 rows with any nonzero raw MagpieData maximum GSmagmom value (max 0.00031), giving a training-fold raw standard deviation of 3.478×10^{-5} for this feature — not exactly zero (so neither TrainFitCompositionScaler's nor fit_standardization's existing exact-zero-variance guards trigger), but $1.1\times10^{4}\times$ smaller than the same feature's full-449-row-population standard deviation (0.392242). Row vtr's raw value for this feature (2.110663, driven by its Fe content) standardizes, against fold 0's training-only mean ($\approx4.1\times10^{-6}$) and std (3.478×10^{-5}), to a z-value of ≈60,683 — the dominant single contributor (41.2% of the squared nearest-neighbor distance; the collinear MagpieData range GSmagmom contributes an identical 41.2%; avg_dev/mean GSmagmom contribute a further 17.7%) to the reported 94,559.5 maximum. A diagnostic sensitivity that drops, per fold, any raw composition-feature dimension whose training-fold-only standard deviation is <5% of that feature's full-population standard deviation (22–26 of 132 dimensions dropped per fold) reduces regime B's maximum standardized distance from 94,559.5 to 26.65 — a 3548× reduction, now consistent with regime B's own 90th percentile (9.16, recomputed; vs. 10.06 undropped) rather than three orders of magnitude beyond it. This confirms the extreme value is a standardization-scale artifact specific to this fold's near-degenerate training-fold variance for magnetic-moment-related Magpie features (itself a direct, expected consequence of Family-GroupKFold removing every magnetic-element-bearing family from that fold's training set) — not a genuine $10^{4}\times$-typical physical composition dissimilarity. It does not change the family-novelty-fraction or Wasserstein-distance qualitative conclusions above, nor any reported number, H1/H2 verdict, or the GO/NO-GO gate (the archived pre-correction record).

Supplementary Note 20. Robust shift-distance correction

This analysis extends an earlier regime-B diagnostic to the full deployment-shift-alignment pipeline (Supplementary Note 19). The threshold was reused unchanged rather than tuned on this output: a raw Magpie feature is dropped from a fold when that training fold's standard deviation is below 5% of the feature's standard deviation across the full 449-row, target-free development population.

Dimensions dropped per fold (of 132 raw Magpie features): regime A, 22/132 in every fold; regime B, 22–26/132 (fold 0, which excludes NASICON together with five smaller families, drops 26 — four more than any other fold); regime C (LOMFO), 22/132 for every one of the 9 major-family exclusions; the external reference fit (fit on the full 449-row pool), 22/132.

Original vs. robust distance distribution:

Cohort	Original max	Robust max	Original q90	Robust q90	Original median	Robust median
A	23.4227	23.4227	4.3568	4.3568	0.2135	0.2135
B	94559.5361	26.6490	10.0632	9.1551	4.7476	4.4295
C	18.5886	18.5886	8.7324	8.7324	4.2561	4.2561
strict cohort	21.6266	21.6266	7.2101	7.2101	3.0114	3.0114
DOI-source-disjoint cohort	21.6266	21.6266	8.2503	8.2503	3.7911	3.7911

Original vs. robust regime-vs-external Wasserstein distance:

Regime	Orig. vs. strict	Robust vs. strict	Orig. vs. DOI-source-disjoint	Robust vs. DOI-source-disjoint
A	2.0998	2.0998	2.9267	2.9267
B	3240.1031	1.3763	3239.5373	1.0095
C	1.0962	1.0962	0.8584	0.8584

Regime B's max drops 3548× (94,559.5 → 26.65) and its Wasserstein distance to the external cohort drops from 3240 (three orders of magnitude beyond every other regime) to 1.4 (the same order of magnitude as regimes A and C), confirming the original value was a standardization-scale artifact of one fold's near-degenerate training-fold variance on magnetic-moment-related features, not a genuine ~10^4×-typical physical composition dissimilarity. **The regime identified as closest to the external cohort by Wasserstein distance is unchanged (LOMFO, both before and after)** — the robust fix changes the scale of one diagnostic number but does not change any qualitative conclusion this study relies on. The original (unmodified) metric is retained in Supplementary Note 19 as the historical/diagnostic record; this section's robust recomputation is the primary sensitivity going forward.

Part F. Risk stratification

Supplementary Note 21. Applicability domain: per-model detail

See Supplementary Note 22 for the finer 16-point-grid AURC recomputation (Fig. 6).

Model	Spearman r	p-value	MAE@100%	MAE@75%	MAE@50%	MAE@25%
Ridge	0.2416	0.0004989	1.3820	1.1684	1.0333	0.9080
Random forest	0.3663	7.117e-08	1.2861	1.0892	0.8636	0.7189
Extremely randomized trees	0.3080	7.39e-06	1.2022	1.0045	0.8459	0.6651
XGBoost	0.3506	2.729e-07	1.3492	1.1595	0.9587	0.7333
CatBoost	0.3012	1.2e-05	1.2916	1.1081	0.9292	0.7014
Shallow MLP	0.3868	1.098e-08	1.4831	1.2415	1.0146	0.8716
Pooled elemental-property model	0.4322	1.077e-10	1.7163	1.3706	1.0400	0.8783
Composition attention	0.3593	1.314e-07	1.2824	1.0214	0.8508	0.7408

AD reference: 449-row OBELiX development population (same population as the external transfer's final fit). Query: 204-row strict external cohort. 8/8 models positive and significant.

Supplementary Note 22. Applicability domain on a 16-point coverage grid

8/8 models show a positive Spearman(AD, |error|) at this finer grid.

Model	Spearman r	p-value	AURC
Ridge	0.2416	0.0004989	1.1313
Random forest	0.3663	7.117e-08	0.9877
Extremely randomized trees	0.3080	7.39e-06	0.9231
XGBoost	0.3506	2.729e-07	1.0565
CatBoost	0.3012	1.2e-05	1.0107
Shallow MLP	0.3868	1.098e-08	1.1415
Pooled elemental-property model	0.4322	1.077e-10	1.2456
Composition attention	0.3593	1.314e-07	0.9709

AURC integrates pooled MAE over the coverage range [0.25, 1.0] (trapezoidal rule, 16-point grid); lower is better. Reported for all 8 models descriptively; no model is chosen based on its own external result.

Supplementary Note 23. Benchmark of risk-stratification comparators

Coverage grid fixed in advance to the applicability-domain analysis's own primary grid (1.0, 0.75, 0.5, 0.25). This analysis did not rerun the external transfer; it reused the authoritative external predictions: seven original frozen models plus the disclosed corrected pooled elemental-property model. Four comparators were benchmarked against the study's existing composition-space AD score (nearest-neighbor distance) on the same risk-coverage/AURC framework: a random-rejection baseline (bootstrap, n_boot=2000, seed=20260921), a Gaussian-KDE source-domain density score (predefined before viewing any result: Scott bandwidth, reference-only PCA to 95% variance for numerical conditioning, see script docstring), a binary/ordinal family-novelty indicator (SEEN=0/AMBIGUOUS=1/UNSEEN=2), and a cross-model prediction-dispersion score (std of the 8 authoritative models' predictions per row — substituted for a literal five-seed dispersion score because the one-shot external transfer never persisted per-row 5-seed std, only its pooled mean/median, and recomputing it would require re-fitting the external transfer's 40 external fits, forbidden this pass).

Mean AURC across 8 models (lower = better risk stratification):

Comparator	Mean AURC	Beats random rejection	Beats family novelty	Positive Spearman(score, |error|)
ad_nn_distance (existing AD score)	1.0543	8/8 models	3/8 models	8/8 models
kde_novelty	1.0954	8/8 models	1/8 models	8/8 models
cross_model_dispersion	0.9657	8/8 models	8/8 models	8/8 models
family_novelty	1.0560	—	—	—
random_rejection	1.3743	—	—	—

Answer to the predeclared main question ("does composition-space NN distance provide risk stratification beyond random rejection, and is its signal merely family novelty?"): the existing AD score clearly beats random rejection (8/8 models, positive Spearman in all 8) — it is a real, non-trivial risk-stratification signal, consistent with the headline result already reported in the main text (Fig. 6). However, its value ADDED beyond simply knowing whether a row's chemistry family was seen in development is weak and inconsistent: the AD score beats the plain family-novelty indicator in only 3 of 8 models (KDE in 1 of 8); in most models, a two/three-level family-seen/unseen flag alone is as good as or better than the continuous composition-distance score at stratifying external risk. Cross-model prediction dispersion — a zero-refit ensemble-disagreement proxy, not the literal five-seed dispersion the task specification named (see substitution note above) — was the strongest comparator overall (lowest mean AURC, beats family novelty in all 8 models), an incidental finding reported

for completeness, not established as this study's headline uncertainty method. Per-model AURC for all five comparators is included in the release (artifacts/authoritative/n21d_ad_benchmark_closure_corrected.json).

Part G. Models and implementation

Supplementary Note 24. Frozen model ladder: hyperparameters and development-split MAE

#	Model (internal key)	Feature schema	n_features	Train MAE	Validation MAE
1	Ridge	Magpie composition (v1)	132	1.1406	1.0646
2	Random forest	Magpie composition (v1)	132	0.3764	0.8549
3	Extremely randomized trees	Magpie composition (v1)	132	0.1506	0.7274
4	XGBoost	Magpie composition (v1)	132	0.2307	0.7019
5	CatBoost	Magpie composition (v1)	132	0.1505	0.7726
6	Shallow MLP	Magpie composition (v1)	132	0.6895	0.8951
7	Pooled elemental-property model	pooled elemental properties (v1)	22	1.3688	1.4472
8	Composition attention	element vocabulary (v1)	vocab=53	0.7310	0.7783

Tuning: Optuna TPE sampler, fixed seed 20260921, 40 trials (sklearn/GBDT) or 20 trials (torch), objective = OBELiX-validation MAE, fit on OBELiX-train only. Multi-seed final-fit list (reserved for the one-shot external transfer): [17, 29, 43, 71, 101].

Supplementary Note 25. Implementation details for reproduction

Consolidates every implementation-level detail needed to reproduce the 8-model ladder from scratch, beyond Supplementary Note 24's summary table.

Optuna search spaces (TPE sampler, fixed seed per fold/final-fit context; every classical/GBDT model gets N_TRIALS=40 trials, every torch model gets TORCH_N_TRIALS=20 trials; objective = validation MAE):

Model	Hyperparameter	Range / choices
Ridge	alpha	log-uniform [1e-3, 1e3]
Random forest	n_estimators	int [100, 600]
Random forest	max_depth	int [2, 20]
Random forest	min_samples_leaf	int [1, 10]
Random forest	max_features	uniform [0.3, 1.0]
Extremely randomized trees	(same 4 as Random forest)	(same ranges)
XGBoost	n_estimators	int [100, 800]
XGBoost	max_depth	int [2, 8]
XGBoost	learning_rate	log-uniform [0.01, 0.3]
XGBoost	subsample	uniform [0.5, 1.0]
XGBoost	colsample_bytree	uniform [0.5, 1.0]
XGBoost	min_child_weight	int [1, 10]
XGBoost	reg_lambda	log-uniform [1e-3, 10.0]
CatBoost	iterations	int [100, 800]
CatBoost	depth	int [2, 8]
CatBoost	learning_rate	log-uniform [0.01, 0.3]
CatBoost	l2_leaf_reg	log-uniform [1e-3, 10.0]
CatBoost	subsample	uniform [0.5, 1.0]
Shallow MLP	n_layers	int [1, 3]
Shallow MLP	hidden_dim (all layers equal width)	categorical {32, 64, 128}

Model	Hyperparameter	Range / choices
Shallow MLP	dropout	uniform [0.0, 0.5]
Shallow MLP	lr	log-uniform [1e-4, 1e-2]
Shallow MLP	weight_decay	log-uniform [1e-6, 1e-2]
Pooled elemental-property model	hidden_dim (single layer)	categorical {16, 32, 64}
Pooled elemental-property model	dropout, lr, weight_decay	same ranges as Shallow MLP
Composition attention	d_model	categorical {16, 32, 64}
Composition attention	n_heads	categorical {2, 4}
Composition attention	n_layers	int [1, 3]
Composition attention	dropout, lr, weight_decay	same ranges as Shallow MLP

Torch architectures. ShallowMLPRegressor (used by both Shallow MLP and, with a single hidden layer, Pooled elemental-property model): Linear -> ReLU -> Dropout blocks stacked n_layers deep at constant width hidden_dim, then a final Linear to a scalar output. CompositionAttentionRegressor (Composition attention): a learned per-element embedding table (nn.Embedding, padding index 0) plus a fraction-value linear projection, both at width d_model; a nn.TransformerEncoder (n_layers TransformerEncoderLayers, n_heads attention heads, dim_feedforward=2*d_model, GELU activation, batch_first=True) refines element representations by attending over the other elements present in the same composition; outputs are fraction-weighted pooled, then passed through a Linear -> ReLU -> Dropout -> Linear head to a scalar. Max elements per composition: MAX_ELEMENTS (fixed constant, see tune_fold.py).

Optimization. Full-batch Adam (torch.optim.Adam, the tuned lr/ weight_decay), MAE loss, up to MAX_EPOCHS=150 epochs, early stopping with PATIENCE=15 epochs of no validation-MAE improvement (best-epoch checkpoint restored). Full-batch training was chosen because every fold/split here has at most a few hundred rows, making per-epoch full-batch gradient steps both simpler and faster than minibatching.

Preprocessing/scaling per model. ridge/Random forest/Extremely randomized trees/ xgboost/catboost/Shallow MLP: TrainFitCompositionScaler (Magpie composition features, Magpie composition (v1), 132 columns; train-fold median-impute then train-fold mean/std standardize, zero-variance columns guarded to std=1.0 — main-text Methods). Pooled elemental-property model: TrainFitElementEmbedder (pooled elemental properties (v1), 22 columns, a train-fold-fit pooled elemental-property embedding, unrelated to the Magpie composition-statistics features used by the other six models). Composition attention: ElementVocabulary (element vocabulary (v1), per-composition element-index/fraction/mask encoding, no continuous-feature standardization — the model learns its own embeddings from indices).

Nested inner/outer split procedure (main-text Methods; the nested internal validation): for every outer fold in every internal-validation regime (A/B/C), hyperparameters are tuned using an 80/20 inner split of THAT fold's own outer-train partition only (inner_early_stop_split, stop_fraction=0.2, same seed convention as outer-fold construction) — the outer-test rows are never seen by inner tuning, and for LOMFO the held-out major family is excluded from outer-train entirely, so it cannot appear in inner tuning either. The final model for that outer fold is then refit on the FULL outer-train partition (inner-train + inner-validation rows) using the inner-tuned hyperparameters, before predicting the outer-test partition.

Pooled MAE vs. macro-family MAE (used throughout Results/SI tables): pooled (micro-average) MAE weights every scored ROW equally, regardless of which chemical family it belongs to (sample_weighted_mae, algebraically identical to computing MAE directly over all rows). Macro-family MAE weights every FAMILY equally regardless of its row count (macro_family_mae, the unweighted mean of each family's own MAE) — a family with 1 row and a family with 130 rows contribute equally to this statistic, making it far more sensitive to a single small family's outlier behavior than pooled MAE is.

Part H. Pre-specification, chronology and provenance

Supplementary Note 26. Pre-specification hash lock

Lock SHA-256: 76510b372bca29624e597aa8f9cffdd7ee32bb33962dff40bb85b41cfa519b98. Re-hashed and matched against the copy recorded by every downstream analysis.

Supplementary Note 27. Chronology of the nested-CV correction, and the pre-submission pooled elemental-property feature-lookup correction

(scripts/transport_analysis/n20_reviewer_remediation.py), reconstructed from the git log of the study branch.

Step	Commit	Date (UTC+6)	Before external visibility?
N3: freeze 8-model hyperparameters	60585f8	2026-09-21T14:43:23	Yes
N4: original internal-CV (later found leaky)	229e5de	2026-09-21T15:21:40	Yes
N5: internal hash lock	600c75c	2026-09-21T15:37:28	Yes
N6: one-shot external transfer (external visibility boundary)	312f0e5	2026-09-21T16:21:16	—
N7: regime-utility comparison (using leaky N4)	a273091	2026-09-21T16:33:21	No
N10: GO/NO-GO decision, GO (using leaky N4)	c73bebb	2026-09-21T17:21:25	No
N11–N13: manuscript draft + four-layer audit (using leaky N4)	1a4bf3b…7eb7f80	2026-09-21T20:01–20:31	No
Discovery + fix of internal-CV hyperparameter leakage (N4b, N7b)	b926660	2026-09-22T01:56:45	No
N9b: 6-model stable sensitivity vs. N4b	5893182 (branch HEAD)	2026-09-22T18:11:00	No

The internal-CV hyperparameter-leakage defect (fixed in b926660) was discovered and fixed **~10.5 hours after** N6's external predictions were generated (312f0e5), and **after** the GO/NO-GO decision, a full manuscript draft, and a prior four-layer audit had already been completed using the leaky N4 numbers. N6's external predictions do not depend on N4's CV output and were **not** rerun for the nested-CV correction; every claim comparing internal-CV MAE to external MAE (H1/H2, regime-utility, GO/NO-GO) **was** recomputed post-fix (N4b/N7b) with an unchanged qualitative verdict.

artifacts/frozen/external_transfer_lock.json is a purely **internal**, local git-committed SHA-256 hash lock plus written protocol text on a private branch — there is no public timestamped registration (no OSF entry, no DOI, no pushed tag on a public remote). This document and the main text use "internally hash-locked protocol" / "pre-specified internal protocol" throughout, never "pre-registered" or "publicly preregistered" .

The post hoc sensitivity analyses, family/duplicate audits, manuscript rebuilds and the final development-only compatibility analysis were performed afterwards on the same branch. None reran the frozen external inference; the only external refit after the primary run is the pooled elemental-property model's disclosed corrective refit (Part H.1 below). The only historical test-label access was the explicitly disclosed full-table compatibility sensitivity, which was replaced by the development-only audit for the final reported compatibility result.

Why the external transfer was not rerun after the nested-CV correction, and what that means for comparing internal and external numbers. The internal-validation regimes (A/B/C, nested per N4b; main-text Methods) and the external transfer (N6) answer two related but procedurally different questions. Nested internal CV re-tunes hyperparameters independently inside every outer fold and reports the resulting out-of-fold error — it estimates the expected performance of the whole tuning-and-fitting PROCEDURE, repeated many times on

different train/test partitions of the same 449-row population. N6, by contrast, evaluates eight FINAL, already-frozen configurations (one hyperparameter set per model, selected once via N3's original train/validation split, before any internal-CV regime or external label was touched) refit once on the full 449-row development population and scored once against the external cohort. These are not the same estimand: nested-CV's internal MAE is a procedure-level estimate with its own fold-to-fold variance (quantified directly in Supplementary Note 16's repeated-seed sensitivity), while N6's external MAE is a single evaluation of one frozen configuration. Rerunning N6 under each new internal-CV variant this study explored would not fix this asymmetry — it would multiply the number of one-shot external evaluations, which the pre-specified hash-locked protocol was explicitly designed to prevent (run_count=1, authorized=True, enforced by artifacts/frozen/external_transfer_lock.json and a guard in scripts/transport_analysis/n6_external_transfer.py that refuses to produce a second primary result). Every internal-vs-external comparison in this study therefore compares a procedure-level internal estimate against a single frozen-configuration external evaluation by construction, not by oversight; we report this asymmetry explicitly rather than treat the two numbers as directly equivalent estimators of the same quantity.

Part H.1. Pre-submission implementation audit: pooled elemental-property model feature-lookup bug. A pre-submission code audit found that features/element_embeddings.py's raw per-element Magpie property lookup called the underlying library function with a plain element-symbol string rather than the pymatgen.core.Element object it requires; every lookup raised an exception that was silently caught and recorded as a missing value, so every one of the pooled elemental-property model's 22 input features was a train-fold-constant zero for every composition, in every split, throughout this study. The model's prediction was correspondingly a near-constant equal to its training-target bias term, which is what the previously reported "near-constant collapse" (Supplementary Fig. 2, original) reflected. The implementation was corrected (the lookup now receives an Element object; a companion imputation-fallback fix and two downstream constant-input guards in evaluation/metrics.py and evaluation/reviewer_remediation.py were added so that a model's constant output is handled as an explicit undefined-correlation case rather than silently producing NaN), and the corrected representation was verified non-degenerate (22/22 usable property columns, 0 zero-variance columns, 362/449 distinct feature rows over the development population, deterministic, train-only-fit, correct unseen-element fallback) before any model was refit. Only this one architecture was affected; the other seven models do not use this code path and are unchanged.

Every internal-validation regime (A–E), the training-size-matched control, the regime-A ten-seed sensitivity, and the one-shot external transfer were rerun for this model only, under the identical frozen protocol (same development population, cohorts, seeds, search space, trial budget, and tuning-seed convention), and every downstream statistic that combines this model's numbers with the other seven (regime-utility tables, the six-model sensitivity, cross-model dispersion, AURC) was recomputed. The external MAE for this model changed only marginally ($1.7159 \rightarrow 1.7163$), because it remains the lowest-ranked model under external transfer either way, but it is no longer a literal constant: it now produces 204 distinct predictions with a real, positive Spearman correlation with the external labels ($\rho=0.39$, previously undefined) alongside a persistent negative R^2 (a systematic scale/offset error, not degeneracy). The six-model sensitivity's one point of apparent support for family-aware CV (a smaller median |MAE gap| for regimes B and C than random CV, 0.38 vs. 0.46) does not survive the correction and is reversed (0.59 and 0.51 vs. 0.46, now favouring random CV, consistent with every other statistic in that subset); LOMFO's rank-agreement in the eight-model primary comparison improves substantially (Spearman ρ $0.19 \rightarrow 0.57$) without changing the overall NOT SUPPORTED verdict for H1/H2. The regime-E (chemical-system-GroupKFold) finding and the eight-model primary regime-utility NOT-SUPPORTED verdicts for H1/H2 are unchanged, but their statistical basis is narrower than previously reported: the primary, row-level B-minus-A and C-minus-A rank-agreement bootstrap CIs (Table 3), which previously excluded zero in random CV's favour, now include zero at this full-cohort, 8-model granularity (they are restored by the LOMFO common-support restriction and are not materially changed by the source/DOI-cluster bootstrap; main text, "Two further checks..."); only the B-minus-A top-model-selection-regret difference remains significant in the primary comparison. The two post hoc Issue 09/10 sensitivities were also recomputed against the corrected 8-model view: Issue 09 Variant B (__MISSING__ rows as singleton groups under Family-GroupKFold) no longer reverses the sign of regime B's rank agreement, but now drives it to a result CI-

distinguishable from random CV's in A's favour (Supplementary Note 34); Issue 10 (formula-balanced external estimand) reconfirms exact numerical identity with the row-weighted estimand under the corrected view, including the formula-cluster bootstrap CIs (above). The corrected values of every affected statistic are in artifacts/authoritative/, which is included in the curated public release. The internal before/after comparison report and the broken pre-correction artifacts are retained unmodified, for provenance, in the authors' internal Git history; they are not included in the curated public release, and no reported number depends on them.

Supplementary Note 28. Figure and table provenance map

Main-text figures (1–6), tables (1–4), Supplementary Figs 1–2 and Supplementary Table 1 map to their generating scripts and source artifacts as follows. All sources are the current results after the pooled elemental-property model correction (Supplementary Note 27); superseded pre-correction results are retained in the authors' internal Git history, are not included in the curated public release and are not used by any figure or table. scripts/manuscript/build_tables.py rebuilds Tables 1–4 from these sources and fails if any number in the manuscript tables differs at the displayed precision; every figure regenerates byte-identically from the listed artifacts.

Item	Script	Source artifact(s)
Fig. 1	scripts/figures/fig1_schematic.py	data/manifests/external_cohort.json, artifacts/authoritative/n4b_nested_internal_validation_corrected.json, artifacts/authoritative/n15_artifacts/n15_source_disjoint.json, artifacts/frozen/model_ladder_frozen.json
Fig. 2	scripts/figures/fig2_external_performance.py	artifacts/authoritative/n15_artifacts/n15_external_uncertainty.json
Fig. 3	scripts/figures/fig3_regime_utility.py	artifacts/authoritative/n7b_regime_utility_nested_corrected.json, artifacts/authoritative/n15_artifacts/n15_stable_model_sensitivity.json, artifacts/authoritative/phase1_factual_fixes_corrected.json
Fig. 4	scripts/figures/fig4_seen_glass_crystalline.py	artifacts/authoritative/phase1_glass_split_corrected.json, artifacts/authoritative/phase2_glass_subset_regime_cis_corrected.json
Fig. 5	scripts/figures/fig5_extrapolation_failure.py	artifacts/authoritative/n4b_nested_internal_validation_corrected.json
Fig. 6	scripts/figures/fig6_risk_coverage_comparison.py	artifacts/authoritative/n21d_ad_benchmark_closure_corrected.json
Supplementary Fig. 1	scripts/figures/si_fig1_shift_alignment.py	artifacts/authoritative/n21e_robust_shift_distance.json (robust diagnostic of Supplementary Note 20; the fold-local reference fix of Supplementary Note 32 leaves every per-fold dimension-drop decision, and therefore this figure, unchanged)
Supplementary Fig. 2	scripts/figures/si_fig2_parity.py	artifacts/authoritative/n6_external_transfer_corrected.json, artifacts/authoritative/n21c_primary_external_context_corrected.json
Table 1	scripts/manuscript/build_tables.py	data/manifests/external_cohort.json, data/manifests/hargreaves_conductivity.json, data/processed/v1/obelix_index.json, artifacts/authoritative/n15_artifacts/n15_source_disjoint.json
Table 2	scripts/manuscript/build_tables.py	artifacts/authoritative/n7b_regime_utility_nested_corrected.json (regimes A–C, eight models), artifacts/authoritative/n15_artifacts/n15_stable_model_sensitivity.json (six models), artifacts/authoritative/phase1_factual_fixes_corrected.json:item3_formula_gkf_utility (regime D)

Item	Script	Source artifact(s)
Table 3	scripts/manuscript/build_tables.py	artifacts/authoritative/ n7b_regime_utility_nested_corrected.json:pairwise_bootstrap_ci (B, C), artifacts/authoritative/phase1_factual_fixes_corrected.json:item4_pairwise_diffs (D)
Table 4	scripts/manuscript/build_tables.py	artifacts/authoritative/ phase1_glass_split_corrected.json:per_model_subset_metrics
Supplementary Table 1	scripts/figures/build_robustness_table.py (build_table4)	artifacts/authoritative/n15_artifacts/ n15_source_disjoint.json, artifacts/authoritative/n15_artifacts/n15_temperature_sensitivity.json
NN-distance forensic diagnostic (not a manuscript figure)	scripts/transport_analysis/ n18_nn_distance_forensic.py	artifacts/authoritative/ n18_nn_distance_forensic.json
Robust shift-distance diagnostic (not a manuscript figure)	scripts/transport_analysis/ n21e_robust_shift_distance.py	artifacts/authoritative/ n21e_robust_shift_distance.json

Every figure script reads only frozen artifacts and writes to manuscript/figures/ (main figures) or manuscript/figures/si/ (Supplementary figures), and is independently re-runnable; all figures are saved as PDF and 300-dpi PNG from the same script invocation. The four main-text table CSVs are written to manuscript/tables/ by scripts/manuscript/build_tables.py. This document (SI) intentionally omits git commit hashes and internal filenames from the main-text narrative; that provenance detail lives here instead. Main-text figure captions and table notes accordingly omit internal artifact "Source: artifacts/..." trailers — every such path already appears in the table above.

Supplementary Note 29. Source records for each Supplementary Note

Source files underlying each Supplementary Note (paths relative to the code repository). The narrative reports under reports/cross_dataset_study/ are retained in the authors' internal Git history and are not included in the curated public release; every listed artifacts/ and data/manifests/ record is included.

SI section	Title	Source file(s)
Supplementary Note 1	External dataset provenance and audit	reports/cross_dataset_study/ n1_external_data_audit.md (full detail).
Supplementary Note 2	403-vs-422 reconciliation	reports/cross_dataset_study/ n1b_403_reconciliation.md (full detail).
Supplementary Note 3	External cohort construction — full stage counts	reports/cross_dataset_study/ n2_external_cohort.md, data/manifests/external_cohort.json.
Supplementary Note 4	Source-only literature overlap (strict cohort vs. OBELiX)	reports/cross_dataset_study/ n2b_source_overlap_audit.md (full detail).
Supplementary Note 24	Frozen model ladder — full hyperparameters and internal MAE	artifacts/frozen/model_ladder_frozen.json, reports/cross_dataset_study/n3_portable_model_benchmark.md.
Supplementary Note 9	Internal validation regimes — nested model selection (primary, nested correction)	reports/cross_dataset_study/ n4b_nested_internal_validation.md, artifacts/authoritative/n4b_nested_internal_validation_corrected.json.
Supplementary Note 26	Pre-specification (hash) lock	artifacts/frozen/external_transfer_lock.json, reports/cross_dataset_study/n5_pretest_lock.md.
Supplementary Note 6	External transfer — full strict and relaxed tables	reports/cross_dataset_study/ n6_external_transfer.md, artifacts/frozen/n6_external_transfer_primary.json.

SI section	Title	Source file(s)
Supplementary Note 10	Validation-regime utility — nested (primary, nested correction)	reports/cross_dataset_study/ n7b_regime_utility_nested.md, artifacts/authoritative/n7b_regime_utility_nested_corrected.json.
Supplementary Note 21	Applicability domain — per-model detail	reports/cross_dataset_study/ n8_applicability_domain.md, artifacts/authoritative/n8_applicability_domain_corrected.json.
Supplementary Note 11	DOI-source-disjoint external sensitivity (full detail)	reports/cross_dataset_study/ n15_reviewer_remediation.md Section 1, artifacts/authoritative/n15_artifacts/n15_source_disjoint.json.
Supplementary Note 19	Deployment-shift alignment (full detail)	reports/cross_dataset_study/ n15_reviewer_remediation.md Section 2, artifacts/authoritative/n15_artifacts/n15_deployment_shift.json.
Supplementary Note 14	Stable-model sensitivity (full detail)	reports/cross_dataset_study/ n15_reviewer_remediation.md Section 3, artifacts/authoritative/n15_artifacts/n15_stable_model_sensitivity.json.
Supplementary Note 8	External uncertainty (full detail)	reports/cross_dataset_study/ n15_reviewer_remediation.md Section 4, artifacts/authoritative/n15_artifacts/n15_external_uncertainty.json.
Supplementary Note 22	AD/AURC, finer 16-point coverage grid (full detail)	reports/cross_dataset_study/ n15_reviewer_remediation.md Section 5, artifacts/authoritative/n15_artifacts/n15_ad_aurc.json.
Supplementary Note 13	Baseline and raw-measurement variability (full detail)	reports/cross_dataset_study/ n15_reviewer_remediation.md Section 6, artifacts/authoritative/n15_artifacts/n15_baseline_variability.json.
Supplementary Note 12	Fixed [20, 30] °C temperature-window sensitivity (full detail)	reports/cross_dataset_study/ n15_reviewer_remediation.md Section 7, artifacts/authoritative/n15_artifacts/n15_temperature_sensitivity.json.
Supplementary Note 17	Family crosswalk and per-model subset detail (full)	artifacts/authoritative/n20_artifacts/ reviewer_family_audit.json
Supplementary Note 15	Exact-duplicate audit and formula-GroupKFold diagnostic (full detail)	artifacts/authoritative/n20_artifacts/ reviewer_formula_duplicate_audit.json
Supplementary Note 5	Paired-formula label-compatibility detail (full)	artifacts/authoritative/ phase2_label_compat_dev_only_full.json
Supplementary Note 27	Nested-CV and provenance timeline	artifacts/authoritative/n20_artifacts/ reviewer_provenance_timeline.json
Supplementary Note 7	Primary external performance context (full detail)	artifacts/authoritative/ n21c_primary_external_context_corrected.json
Supplementary Note 23	Applicability-domain benchmark closure (full detail)	artifacts/authoritative/ n21d_ad_benchmark_closure_corrected.json
Supplementary Note 20	Robust shift-distance fix (full detail)	artifacts/authoritative/ n21e_robust_shift_distance.json
Supplementary Note 16	Repeated random-5-fold CV fold-randomness sensitivity (full detail)	artifacts/authoritative/ n21_repeated_cv_sensitivity_corrected.json
Supplementary Note 18	UNSEEN-family material-class refinement: glass/glass-ceramic vs. crystalline (full detail)	artifacts/authoritative/ phase1_predeclared_corrected.json, artifacts/authoritative/phase1_glass_split_corrected.json

Part I. Additional post hoc robustness controls

Seven additional post hoc robustness controls were run for this submission, each specified in a written, hashed plan (reports/remediation/specs/) before any number was computed, none rerunning the frozen external transfer or opening the frozen 113-row OBELiX test split. The specifications and their SHA-256 hashes (reports/remediation/specs/, HASHES.txt), the scripts (scripts/analysis/issue*.py) and the result records (artifacts/authoritative/issue*.json) are included in the release.

Supplementary Note 30. Common-support LOMFO sensitivity

LOMFO (regime C) scores only the 396 development-population rows belonging to a major family (n>=15); regimes A and B score all 449. Restricting regimes A and B's internal pooled MAE to this identical 396-row common support (row-count-weighted recombination of already-frozen per-family MAE; no retraining) gives, for the corrected 8-model view: A's median |MAE gap| 0.6106 (vs. 0.4531 full-support), B's 0.6397 (vs. 0.6810 full-support) -- the point estimate still favors A over B on this restricted population, as it does under full support, but by a much narrower margin (0.61 vs. 0.64, against 0.45 vs. 0.68 under full support); the 95% bootstrap CI on the B-minus-A delta includes zero ([-0.2952, 0.3655]). Rank agreement moves the other way: under full support, the corrected B-minus-A Spearman and Kendall differences no longer exclude zero (Supplementary Note 27), but restricting to the common support restores a result that excludes zero in A's favour (Spearman rho delta -0.2857, 95% CI [-0.5476, -0.0952]; Kendall tau delta -0.3571, 95% CI [-0.6429, -0.0714]); regret is unchanged (delta +0.1471, 95% CI [0.0337, 0.2638], excluding zero both times). **Conclusion: on the corrected 8-model view, the common-support restriction no longer flips the median-|MAE-gap| point estimate to favor B -- it still favors A, just more narrowly -- while it restores statistical significance for the rank-agreement comparisons that the primary full-cohort analysis lost after the pooled elemental-property model's representation was corrected (Supplementary Note 27); regret is robust to the restriction either way.**

Supplementary Note 31. Training-size-matched control for Family-GroupKFold

Regime B's fold 0 removes NASICON plus five smaller families (213/449 = 47.4% of the pool) from training, confounding training-size loss with family-exclusion distribution shift. A plain random row-level holdout of the identical per-fold test size (ignoring family membership; 3 deterministic seeds, full nested per-fold retuning identical to the primary protocol) stays within the same order of magnitude as regime A for every model, including the two that catastrophically blow up under regime B itself: ridge (A = 1.410; B = 143.129; size-matched control = 1.410, 1.299, 1.281 across the 3 seeds) and the shallow MLP (A = 1.029; B = 969.424; size-matched control = 1.154, 1.153, 1.137). **This directly confirms the manuscript's mechanism claim**: the extrapolation failure is caused by removing an entire chemical family's feature distribution from training, not by training-set size reduction alone. Every other model shows only modest degradation under the matched-size control, comparable to its ordinary degradation under regime B.

Supplementary Note 32. Strictly fold-local robust shift-distance correction

Supplementary Note 20's robust shift-distance fix used, as its near-degeneracy-guard denominator, each feature's standard deviation over the full 449-row development population -- a population that includes each fold's own held-out rows' feature values (never labels; the diagnostic remains fully target-free throughout). A corrected version instead uses, per fold, the median of every *other* fold's own train-only standard deviation as the reference (a perfectly fold-blind reference is not constructible for a K-fold partition of a fixed pool -- every row is, by construction, a training-set member of every fold except the one where it is held out, so any non-trivial population-style reference necessarily touches it somewhere; this construction removes the specific problem that a fold's own unusually large exclusion, such as regime B's fold 0, no longer single-handedly determines the threshold it is judged against). Checked directly, not assumed: the corrected reference produces **identical per-fold dimension-drop decisions** to the original (regime A: 22/132 dropped in every fold under both methods; regime B: 26, 22, 23, 22, 22 dropped per fold under both methods), so every downstream

distance, Wasserstein value, and the closest-regime-to-external verdict (LOMFO, unchanged) are numerically identical. The audit trail is now correct; no manuscript number changes as a result.

Supplementary Note 33. Source/DOI-cluster bootstrap

The primary and Supplementary Note 8 uncertainty analyses resample external rows independently. Connected-component clustering over the strict cohort's formula-DOI bipartite graph (213 pre-aggregation retained rows, 204 formulas, 91 DOIs, reproducing Supplementary Note 4's counts exactly) yields 85 source clusters (largest = 10 rows). Resampling clusters instead of rows (2,000 iterations, same fixed seed) widens every model's external-MAE 95% CI, as expected (for example, extremely randomized trees [0.978, 1.464] at cluster level versus [1.032, 1.385] under this analysis's own row-level resampling). It changes no regime comparison's zero-exclusion status. After the pre-submission correction of the pooled elemental-property model (Supplementary Note 27), the row-level B-minus-A and C-minus-A rank-agreement CIs already include zero (B-minus-A Spearman rho -0.0952, row-level [-0.3339, 0.0476], cluster-level [-0.4048, 0.1190]; B-minus-A Kendall tau -0.1429, row-level [-0.4286, 0.0000], cluster-level [-0.4286, 0.0000]; C-minus-A Spearman rho -0.0714, row-level [-0.6190, 0.1667], cluster-level [-0.6429, 0.2619]; C-minus-A Kendall tau -0.0714, row-level [-0.5000, 0.0714], cluster-level [-0.6429, 0.2143]), and they remain so under clustering. The median-|MAE-gap| differences include zero under both resampling units, and the B-minus-A regret difference (0.1471) excludes zero under both (row-level [0.0337, 0.2638]; cluster-level [0.0052, 0.3122]). Point estimates are identical by construction. An earlier version of this analysis, computed before the pooled elemental-property model was corrected, found that clustering moved four rank-agreement CIs from excluding to including zero; that finding does not survive the correction, because the corrected row-level intervals already include zero. Same-source dependence still widens every interval, so the row-level CIs should be read as a lower bound on external-cohort uncertainty.

Supplementary Note 34. __MISSING__-family sensitivity

18 of 449 development-population rows lack a Family label and are filled with the __MISSING__ sentinel, which is then treated as its own group for regimes B/C. Both variants below were rerun for the pooled elemental-property model only, after the pre-submission feature-lookup correction (Methods; Supplementary Note 27); the other seven models' numbers are unchanged from the historical analysis. **Variant A (exclude these 18 rows from evaluation; row-count-weighted recombination of already-frozen per-family MAE, no retraining):** regime-utility rank agreement and regret are **exactly unchanged** for both B-minus-A and C-minus-A (identical Spearman rho, Kendall tau, and regret, to four decimal places, with and without __MISSING__); the median-|MAE-gap| point estimate shifts modestly (B-minus-A: 0.2278 to 0.2888; C-minus-A: 0.0618 to 0.1361) but neither crosses the excludes-zero/includes-zero boundary in either direction. **Conclusion for Variant A: the primary result does not depend on how the 18 __MISSING__ rows are handled in evaluation.**

Variant B (treat each __MISSING__ row as its own singleton group under Family-GroupKFold; genuine nested-CV rerun, regime B'): the catastrophic blowup essentially disappears for both previously-unstable models (ridge 143.129 to 2.333; shallow MLP 969.424 to 1.763), while every other model shows a modest *increase* relative to regime B (e.g. catboost 1.487 to 1.842; pooled elemental-property model 2.702 to 2.688, essentially flat). Regime B's Spearman rank agreement with external ranking is already weak in the corrected primary comparison (0.5476, no longer significantly different from random CV; Table 3) and **falls further under B' (0.1905)**, and Kendall tau likewise falls from 0.3571 to 0.1429; the B'-minus-A rank-agreement 95% CIs clearly exclude zero in A's favor (Spearman [-0.7857, -0.1429]; Kendall [-0.6429, -0.0714]) even though neither B-minus-A difference does (Table 3) -- unlike an earlier analysis affected by the since-corrected feature-pipeline bug, B' no longer reverses the *sign* of B's rank agreement (both stay positive: $0.55 \rightarrow 0.19$ and $0.36 \rightarrow 0.14$), it only drives an already-non-significant positive association to a significantly worse one. Median |MAE gap| (B-minus-A: 0.2278 to 0.2834, CI still includes zero) and regret (unchanged at 0.1471, CI excludes zero both times) are not materially changed in direction. **Conclusion for Variant B: H1/H2 remain NOT SUPPORTED under this alternative grouping (no new support for family-aware validation), but this reveals a genuine fragility -- regime B's weak positive rank-agreement point estimate is not robust to an**

arbitrary, defensible implementation choice for the __MISSING__ sentinel group, and degrades to a result statistically distinguishable from random CV's, in A's favor. The primary comparison throughout this study uses the original, single, pre-specified __MISSING__-as-one-group convention, unchanged; this sensitivity is reported as an added limitation, not substituted into the headline result.

Supplementary Note 35. Chemical-system-GroupKFold sensitivity (regime E)

The exact-formula-disjoint control (regime D) removes only exact reduced-formula duplication. This note extends the control to the pymatgen element-set ("chemical system") level, following Supplementary Note 15's observation that 63 chemical systems (334 rows) share an element set across more than one distinct stoichiometry. 153 unique chemical systems span the 449-row development population (largest group 4.5% of the pool), so, unlike Family-GroupKFold, no single fold can strip an entire feature region from training; a genuine nested-CV rerun under this grouping (regime E) confirms this: the largest E-vs-A pooled-MAE ratio is 1.30x (composition attention), against up to 941.8x under regime B.

Regime E's regime-utility statistics are a genuine, unanticipated finding, reported as found. E's median |MAE gap| (0.1045) is substantially smaller than A's (0.4531), with a 95% CI clearly excluding zero in E's favor (delta -0.3486, 95% CI [-0.4263, -0.1370]) -- chemical-system-GroupKFold's internal error estimate is closer in *magnitude* to the external MAE than random CV's. Mechanically, E's internal MAEs are modestly higher than A's for every model (e.g. ridge 1.41 to 1.56, extra trees 0.75 to 1.19), because E's fold construction removes a mild form of composition redundancy that random CV does not control for, landing closer to the external MAE's ~1.2-1.8 range. Rank agreement is essentially tied with A (Spearman rho exactly equal, delta 0.0, 95% CI including zero, [-0.1667, 0.0714]; Kendall tau deterministically slightly lower for E, delta -0.0714, 95% CI [-0.0714, -0.0714]), and regret is worse for E (delta +0.0895, 95% CI [0.0019, 0.1726], excluding zero in A's favor).

This is not evidence that chemistry-family-awareness helps -- regimes B/C already answer that question in the negative -- because chemical systems are a much narrower, more numerous grouping (153 groups, largest 4.5% of the pool) than chemistry families (as few as 9 major groups, up to 47.4% of the pool in one fold); this is a different, weaker redundancy-control axis, not a stronger family-awareness test. The honest reading: some degree of composition-redundancy control beyond exact-formula, short of full family exclusion, improves internal-external MAE-magnitude calibration without the extrapolation-failure cost family exclusion causes, but does not improve, and slightly worsens, model-selection reliability.

Supplementary Note 36. Formula-balanced external-estimand sensitivity and formula-cluster bootstrap

Spec: reports/remediation/specs/issue10_formula_balanced_sensitivity.md, hashed before any number below was computed (reports/remediation/specs/HASHES.txt). **Analysis script:** scripts/analysis/issue10_formula_balanced_sensitivity.py (historical 7/8-model run; superseded for the pooled elemental-property model's row by scripts/analysis/corrective01_issue10_formula_balanced_sensitivity.py after Supplementary Note 27's correction). **Artifact:** artifacts/historical/issue10_formula_balanced_sensitivity.json (historical; retained in internal Git history, not included in the curated public release); artifacts/authoritative/issue10_formula_balanced_sensitivity_corrected.json (corrected 8-model view, authoritative for the numbers below).

Motivation. The primary external metrics assign equal weight to the strict Hargreaves cohort's aggregated formula-level records. As a post hoc identity check, this note asks whether an explicit equal-weight formula grouping -- the estimand that would be needed if several retained rows had instead shared one reduced formula -- changes external MAE or model ranking for the frozen 204-record strict cohort actually used by the one-shot external transfer (artifacts/frozen/n6_external_transfer_primary.json).

Premise check. aggregate_by_formula (src/sse_transfer/data/hargreaves_cohort.py) builds the strict cohort by grouping the 213 pre-aggregation retained rows by reduced_formula and emitting one aggregated formula-level record per group, using the unweighted median target within each group, *before* N6 ever reads a target value. The analysis script verifies this with collections.Counter, rather than assuming it or aborting on repeats: it

computes n_unique_reduced_formulas, n_repeated_formula_groups (formula identities with count > 1) and n_duplicate_excess_rows (total rows minus unique formulas) directly from the frozen cohort file, and separately verifies -- by formula multiset, not by requiring uniqueness -- that each of the eight models' frozen predictions covers the identical formula population, so residuals and formulas cannot silently misalign even if a future cohort does contain repeats. For this frozen cohort: 204 rows, 204 unique reduced_formula values, **zero repeated-formula groups**, zero duplicate-excess rows. Every model's frozen prediction set independently confirms the same 204-formula multiset.

Formula-balanced MAE. Defined as MAE_formula,m = mean_f(mean_{i ∈ f}|y_i − ŷ_{i,m}|), implemented as an explicit group-then-average (verified on synthetic multi-member-group data in tests/unit/test_regime_utility.py to equal manual group averaging, and to differ from the plain row mean whenever group sizes are unequal). Given the premise check above, this is numerically identical to the row-weighted MAE for every model:

Model	Row-weighted MAE	Formula-balanced MAE
extra_trees	1.2022	1.2022
composition_attention	1.2824	1.2824
random_forest	1.2861	1.2861
catboost	1.2916	1.2916
xgboost	1.3492	1.3492
ridge	1.3820	1.3820
shallow_mlp	1.4831	1.4831
pretrained_element_embedding	1.7163	1.7163

The formula-balanced model ranking is identical to the row-weighted ranking (Fig. 2) for all eight models. The pooled elemental-property model's row above reflects its corrected representation (Supplementary Note 27). This note's own script (scripts/analysis/issue10_formula_balanced_sensitivity.py) correctly refuses to run against the corrected, non-primary external-transfer view (its authorization guard is designed to reject exactly this substitution); a separate script reusing the identical formulas and bootstrap procedure (scripts/analysis/corrective01_issue10_formula_balanced_sensitivity.py) recomputed every statistic below, including the formula-cluster bootstrap CIs, against the corrected eight-model view, without touching that guard or the historical one-shot lock.

Regime-utility recomputation. Feeding the formula-balanced external MAE into the same regime_summary function used for the primary comparison gives, for regimes A, B, C and D (8-model and 6-model subsets), statistics identical to the already-reported row-weighted values (median |gap|, Spearman ρ, Kendall τ, and regret all match to floating-point precision; e.g. A: 0.4531/0.6429/0.5000/0.0000, D: 0.4009/0.6429/0.5000/0.0000, matching the main-text and Supplementary Note 15 values exactly).

Formula-cluster bootstrap. Resampling unique formula groups (not rows) with replacement, 2,000 iterations, the same fixed seed (20260921) used by every other bootstrap in this study, gives external-MAE 95% CIs for all eight models and the regime-utility pairwise deltas B-minus-A, C-minus-A (from the corrected artifacts/authoritative/n7b_regime_utility_nested_corrected.json) and D-minus-A (from the corrected post hoc regime-D bootstrap in artifacts/authoritative/phase1_factual_fixes_corrected.json's item4_pairwise_diffs.formulaGKF_minus_A). The script hard-checks each pairwise statistic's point delta against its pre-existing row-level reference to 1e-9 tolerance and raises if violated, because point-delta agreement is a required scientific invariant (given the premise check, a cluster bootstrap over singleton formula groups draws the same multiset of rows, with the same multiplicities, as a row bootstrap over individual rows, for any correct implementation of either). CI-endpoint agreement with the row-level reference is computed and reported rather than asserted, because the spec identifies a cluster bootstrap over singleton groups (rng.choice) and a row-level bootstrap over individual rows (rng.integers) as different pseudorandom draw sequences over the same distributional target, so a CI difference would reflect sampling noise, not a population-weighting

effect. For this run, every one of the twelve pairwise point deltas (three pairs by four statistics) matched its reference to 1e-9, and every corresponding CI endpoint also matched exactly (zero observed difference) -- an empirically observed implementation-level identity, not a mathematical requirement, consistent with Generator.choice(a, size=n, replace=True) over a = arange(n) happening to call Generator.integers(0, n, size=n) internally under the current NumPy version; that bit-for-bit coincidence is a NumPy implementation detail, not a documented API guarantee.

Decision. External model ordering: unchanged. Validation-regime-utility magnitude and rank-agreement calibration: unchanged. Selection-regret conclusions: unchanged. The paper's deployment-conditional central claim: unaffected. Repeated-formula weighting is therefore not a vulnerability of the external estimate reported in this manuscript, because the cohort-construction pipeline already guarantees row and formula granularity coincide; this is reported as a verified finding rather than an assumption, and the check would need to be redone, not presumed, if a future external cohort were built without row-per-formula aggregation.

Supplementary References

S1. Therrien, F. et al. OBELiX: a curated dataset of crystal structures and experimentally measured ionic conductivities for lithium solid-state electrolytes. Digit. Discov. **5**, 910–918 (2026).

S2. Hargreaves, C. J. et al. A database of experimentally measured lithium solid electrolyte conductivities evaluated with machine learning. npj Comput. Mater. **9**, 9 (2023).